



Frontier AI for Actuaries.

AI helps actuaries perform complex, multi-step actuarial tasks with every step documented and reproducible.

THE WORKBENCH

One collaborative workspace for pricing, reserving, and filing

For all actuaries at admitted and specialty carriers, MGAs, and reinsurers.

August 2026
CONFIDENTIAL

Actuarial work is not one workflow. It is a web of intricate, connected tasks.

The more steps an actuary delegates to AI, the less certain the output. Tesora traces and documents every step, so the actuary stays in control.



Philo Bishay  • 1st

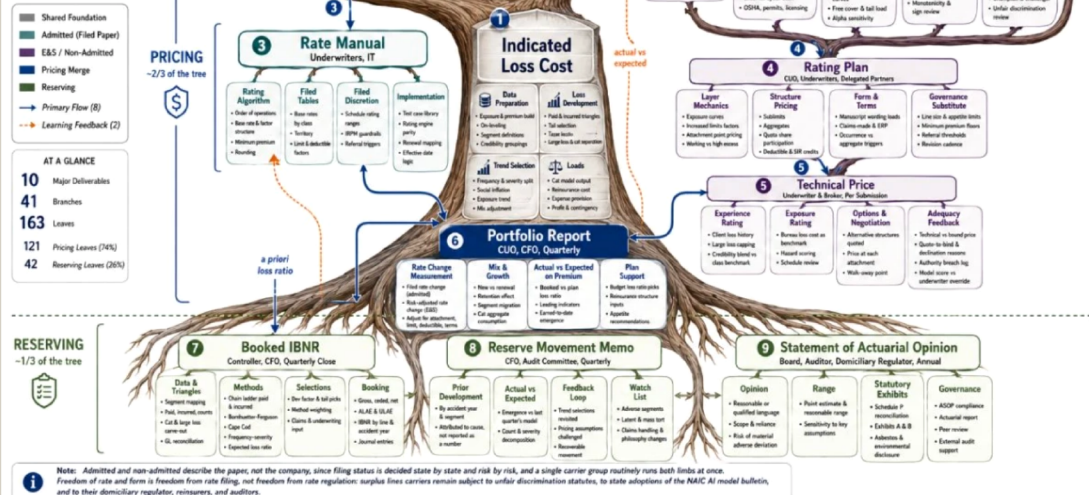
Cofounder & Head of Product @ Tesora | Actuarial | Wharton MBA

2w •

Every actuary dreads one question, "What do actuaries do?" I am happy to say we are no closer to an answer, but if you want to avoid further questions, send this:

What Actuaries Deliver

Two questions. All decisions.
What do we charge?
What do we already owe?
 Actuaries own both.

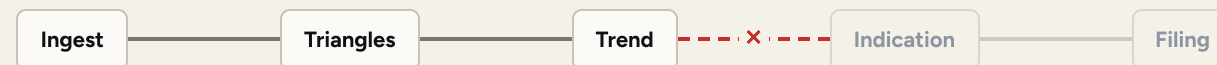


Over 150 workflows are core to the actuarial function



■ 121 pricing ■ 42 reserving

one mark per workflow



One weak handoff and everything downstream is unusable.

One shot, and you get a number nobody can trace.

- A rate filing or a booked reserve needs dozens of these correct, in sequence, each one feeding the next.
- Automate a single step and the chain still breaks at the handoff after it.



One agentic harness across all of it

A system of agents keeps every task linked and every value cited, from ingest to filing. That orchestration is the hard part to build, and the harder part to replicate.



No single agent does the whole job, and **none of it reaches you unchecked.**

We watched how actuaries actually do the work, mapped it into sub-agents, and wrote evals for each one, with audit agents sitting behind them. In a domain this intricate, that is what delivers a business outcome without a hallucinated number or a botched handoff between steps.

A REQUEST ARRIVES

1

The coordinator breaks the request into tracked work

It picks the specialists, runs them in parallel, and reports back.

2

A specialist agent owns each domain end to end

Each returns a finished deliverable in the format that domain expects.



Pricing and experience

Loss ratios, trend, and frequency-severity, as a documented report.



Reserving

Loss development, IBNR, and triangles, as a formatted exhibit.



Rating models

Builds, validates, and tests a rater from its filing.



Document extraction

Filings, loss runs, and PDFs into clean tables, cell by cell.

Audit your existing work

Reads a spreadsheet, document, or formula you already have and reports what is wrong.

3

Review agents check the work before you do

Read access only. They report findings, and never edit the work.

Actuarial expert

Method questions, answered from the standards library.

Citation check

Every figure hyperlinks to its page. No link, no value.

Value tracing

Each number ties to a document, or to a sum of ones that do.

Coverage check

Reads what was not cited, to catch anything skipped.

YOUR ACTUARY REVIEWS AND SIGNS OFF

Each step runs on the model that suits it. Extraction runs small and judgment runs strong, so cost tracks the actuarial outcome rather than the token count.



Rate changes in **days**, not months.

Source a rater from a SERFF filing, an Excel file, or a script. Tesora rebuilds it, tests it, and deploys it whenever underwriting is ready.

BEFORE: BY HAND

4-6 months

From a requested change to a filed, live rater, and where the months actually go

- | | | |
|----|---|-------|
| 01 | Spec the change
Actuary writes up the revised rates and factors | DAYS |
| 02 | IT ticket, then the queue
The change waits for developer capacity | WEEKS |
| 03 | Re-code the rater
Engineering reimplements the logic by hand | WEEKS |
| 04 | Rebuild exhibits, 50 states
Each state reformatted by hand for SERFF | WEEKS |
| 05 | Manual QA
No test suite, so regressions are caught by eye | WEEKS |
| 06 | Compliance sign-off
Actuarial and legal review the filing | DAYS |
| 07 | File and answer objections
Back and forth with each department of insurance | WEEKS |
| 08 | Release window
Deploy only in the next scheduled window | WEEKS |

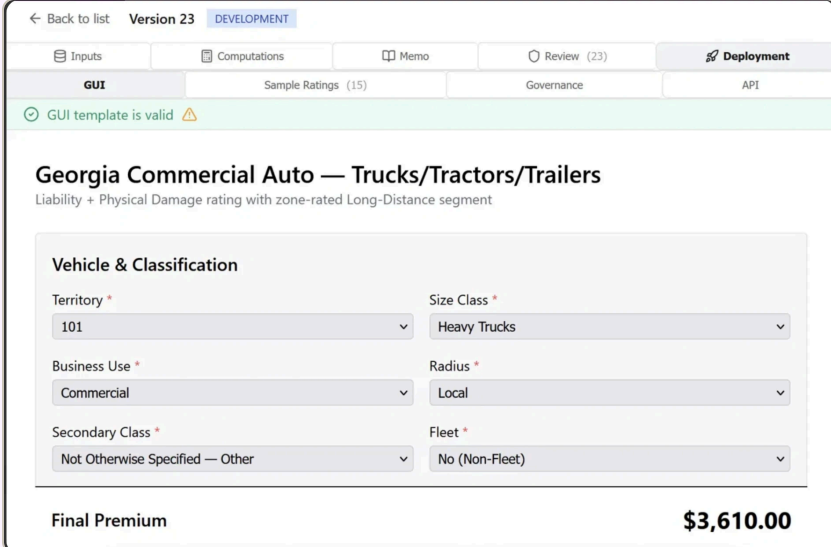
AFTER: Tesora

1-2 days

Deploy to a GUI, an API, or back into Excel.

You give: a plain-language change and the source rater, from SERFF, Excel, or a script.

You get: a tested rater with a working GUI, sample ratings, a versioned API, or an Excel file your team can open.



The screenshot shows a web application interface for rate calculation. At the top, there's a navigation bar with 'Back to list', 'Version 23', and a 'DEVELOPMENT' status. Below this are tabs for 'Inputs', 'Computations', 'Memo', 'Review (23)', and 'Deployment'. The 'Inputs' tab is active, showing a 'GUI' section with a 'Sample Ratings (15)' and 'Governance' sub-sections. A green status bar indicates 'GUI template is valid'. The main content area is titled 'Georgia Commercial Auto — Trucks/Tractors/Trailers' with a subtitle 'Liability + Physical Damage rating with zone-rated Long-Distance segment'. Below this is a 'Vehicle & Classification' section with several dropdown menus: 'Territory' (101), 'Size Class' (Heavy Trucks), 'Business Use' (Commercial), 'Radius' (Local), 'Secondary Class' (Not Otherwise Specified — Other), and 'Fleet' (No (Non-Fleet)). At the bottom, the 'Final Premium' is displayed as '\$3,610.00'.



The triangle is squared before you sit down.

A quarterly close moves between actuarial, IT, and finance, and most of the calendar goes to assembling the data rather than judging it.

BEFORE: BY HAND

~2 weeks

From close to a booked IBNR, and who is waiting on whom

- | | | |
|---|-----------------------------------|-------|
| 01 | Request the data | HOURS |
| Actuarial files a ticket, and IT owns the queue | | |
| 02 | Wait on the extract | DAYS |
| The pull sits behind other reporting work | | |
| 03 | Reconcile to the ledger | DAYS |
| Paid and incurred tied back to finance, by hand | | |
| 04 | Rebuild the triangles | DAYS |
| Segment mapping and large-loss carve-outs redone each quarter | | |
| 05 | Select factors and methods | DAYS |
| Chain ladder, Bornhuetter-Ferguson, Cape Cod, tail | | |
| 06 | Rebuild the exhibits | DAYS |
| Roll-forward and movement memo reformatted by hand | | |
| 07 | Peer review | DAYS |
| A second actuary retraces the work from scratch | | |
| 08 | Book and explain | HOURS |
| Answer the CFO, the auditor, and the board | | |

AFTER: Tesora

3-4 hours

Triangles squared on ingest, so the quarter starts at the selection

You give: the loss run, as it comes out of the system.

You get: reconciled triangles, method comparisons, and an IBNR with every value traced to source.





Rebuild what a competitor charges, then **price your book through it.**

A filing gives you tables, rules, and inputs rather than a working model, and the order of operations is rarely spelled out. Turning that into something you can run your own risks through is the work.

BEFORE: REBUILT BY HAND FROM SERFF

~1 week

From a competitor filing to a usable read, and where it goes wrong

- | | | |
|----|--|---------|
| 01 | Pull the filings
An analyst searches SERFF, state by state | HOURS |
| 02 | Read the rate manual
Rules, factors, and order of operations, by eye | DAYS |
| 03 | Rekey the rate tables
Hundreds of tables retyped into a spreadsheet | DAYS |
| 04 | Rebuild the algorithm
The logic is inferred, then argued over | DAYS |
| 05 | Price a sample
A dozen test risks, never the real book | HOURS |
| 06 | Compare by hand
Deltas built in a one-off workbook nobody owns | DAYS |
| 07 | Explain the gaps
Each difference traced back to a single factor | DAYS |
| 08 | The read goes stale
They filed again while you were rebuilding | ONGOING |

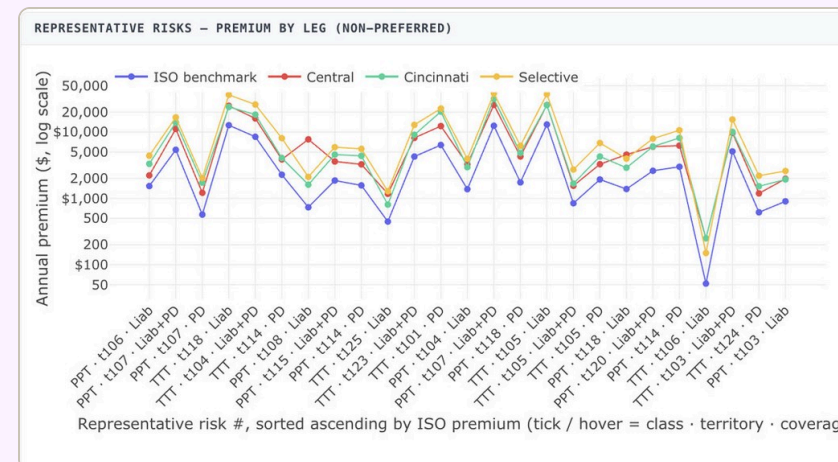
AFTER: Tesora

10-15 minutes

An executable competitor rater, and your whole book priced through it

You give: a competitor, a state, and a line. Or the SERFF filing itself.

You get: their rating logic running against your risks, every factor cited to the page it came from.



Built from public rate filings, and any additional data sources you provide.

The full context is baked into the work.

Every session holds the data it used, the assumptions behind it, and the reasoning that got there. Reopen it years later and nothing has to be reconstructed, because the context was never separate from the answer.

Tesora

Home

Work Tracking

ARTIFACTS

Folders

Reports

Raters

Routines

APPS

Compute St...

Underwriting

Rater Recon...

Competitive ...

GLM Studio

Triangle Bull...

All Apps

Support

Settings

vivek

Back

EXAMPLE Bernegger 2024 — Integrated experience/exposure rating (ASTIN)

EX-VLT-bernegger-2024-integrated-rating

Analyze Data

Build

Rate Rule Form

Profit & Contingencies

7.0%

See previous prompt

Total expense & profit load

47.0%

Expected Loss & LAE Ratio (ELR)

53.0%

Projected Ultimate Loss Ratio

54.7%

Adequacy margin (ULR – ELR)

1.7%

short

Read: the thin spot is the 47% load (30 pts of it commission), and the book runs ~1.7 pts past its permissible loss ratio — a modest inadequacy with no cushion.

Caveat (carried from the exhibit): ILFs are relativities, not dollar prices; no competitor base rates or retention/deductible tables exist in the folder. Preliminary analyst work product, subject to credentialed-actuary review.

The full CSV is downloaded at [CIC_vs_Peers_ILF_Competitive_Benchmark.csv](#), and the narrative report is [CIC vs Cyber Peers — ILF & Competitive Benchmark ...](#). Want me to render this as a styled Excel exhibit, or open the report next?

Back to documents

bernegger-2024-integrated-experience-exposure-rating.pdf

Download

Origin

User upload

Type

application/pdf

Size

1.5 MB

Status

pending

Uploaded

Aug 7, 2026, 06:57 PM

Document ID

DOC-5c0c8dfe-1a3a-4eeb-8c76-5ab1947ac4dd

File ID

54238c79-b072-45c7-92cb-096cfb81f242

SUMMARY

No summary available. The ingestion workflow either hasn't finished or didn't produce one for this document.

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Generic framework for a coherent integration of experience and exposure rating in reinsurance

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Keywords: Experience rating; exposure rating; short tail; long tail; generative model; Bayesian inference; reduced statistics; model calibration; Monte Carlo simulation

Abstract

This article introduces a comprehensive framework that effectively combines experience rating and exposure rating approaches in reinsurance for both short-tail and long-tail businesses. The generic framework applies to all suitable lines of business and produces compelling nonproportional reinsurance. The approach is based on three pillars that enable a coherent usage of all available information. The first pillar comprises an exposure-based generative model that emulates the generative process leading to the observed claims experience. The second pillar encompasses a standardized reduction procedure that maps each high-dimensional claim object to a low-dimensionally reduced random variable. The third pillar comprises calibrating the generative model with retrospective Bayesian inference. The derived calibration parameters are fed back into the generative model, and the reinsurance contracts covering future cover periods are used by projecting the calibrated generative model to the cover period and applying the future contract terms.

1. Introduction

There are two approaches to probability theory: the first is the frequentist interpretation, also known as standard, orthodox, or classical statistics, and the second is the Bayesian interpretation (Jaykin et al., 1994; Lee, 2012; Gelman, 2013; Klugman et al., 2019; Murphy, 2022, 2023). This distinction also applies to the various statistical and modeling approaches.

The Bayesian interpretation has multiple advantages. Not frequentist approaches are still popular and widely used for two main reasons. First, they are computationally 'cheap' compared to Bayesian methods, and second, they do not presuppose knowledge regarding priors. Bayesian inference is, nevertheless, widely applied in various domains of science (Pear et al., 2022; Palmer, 2022; Harrison et al., 2023), and it has become the foundation of machine learning (Murphy, 2022, 2023). Bayesian approaches are also gaining popularity in disciplines like insurance and reinsurance, which have traditionally been dominated by frequentist approaches (Bühlmann and Straub, 1970; Bühlmann and Gisler, 2005; Zhang et al., 2012; Klugman et al., 2019; Geffard et al., 2023).

Reinsurance companies face multiple challenges when using models for assessing the assumed risks and for taking business decisions like risk selection, transactional pricing, reserving, investing, and corporate risk management, and they rely on a variety of methods and tools for managing these risks (Antal, 2009; Albrecher, 2017; Mühlenthal, 2022). The actuarial models implemented by a reinsurer are based on simplified assumptions, and reinsurers know that the underlying assumptions are rarely fulfilled in the real world. One way to deal with the issue is to complement experience-based rating models with exposure-based models. The latter can incorporate all relevant features of the generative process that affect the rating. All such features must be calibrated, which is accomplished with market data. However, cedent-specific features (hidden variables) must also be reflected in the rating of reinsurance contracts.

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Implementation

The models are wired into your loss runs, policy systems, and ledger, running inside your own environment.

Governance

Author, reviewer, and approver are separate identities, and the trail survives every promotion. Finance and IT find what they need without waiting on an actuary.

Compliance

Built to ASOP from the start, and SOX-aligned for carriers with public-company parents. Same input, same answer, every time.

Continuity

When an actuary leaves, their assumptions, sources, and reasoning stay in the session. The next person picks up the thread.

Every request from the platform pings our chief actuary and CTO. Support is 24/7, agents and people together, so you are never left with a bot. A real person answers within two to three minutes, and we have had no unplanned downtime to date, which is **model routing** doing its job.



What's the best way to **implement AI**?

Your implementation of AI today will dictate its value, translatability, and reproducibility going forward.

	Actuarial consultants Milliman, WTW, Oliver Wyman	Your team with AI In-house, on ChatGPT or Claude	Actuarial software hyperexponential, Earnix, Akur8	 Tesora Frontier models, in your format
Turnaround	☐ Weeks to months, in their queue	☒ Quick drafts, still checked by hand	☒ Fast and repeatable once it is live	☒ Minutes to days, in your workflow
Setup & fit	☒ A fresh engagement scoped each time	☒ No one knows your book better	☒ Heavy implementation on their platform	☒ 60-day build in your VPC, on your own data
Your raters & filings	☐ Generic deliverables in their template	☒ Your format, rebuilt by hand each time	☒ Rebuilt inside their system, their way	☒ Your raters, versioned and ready to deploy
Source & audit	☒ Independent, credible workpapers	☐ No source to trace, and the answer drifts	☒ Logged inside the platform	☒ Cell-level citations, append-only log
Scope	☒ Whatever you scope and pay for	☒ One step at a time, gaps in between	☒ Pricing or reserving, rarely filing too	☒ Pricing, reserving, filing, and close in one place

☒ Full capability ☐ Gap



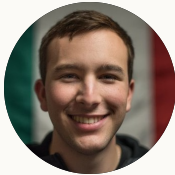
A team built at the intersection of frontier AI research and decades of actuarial experience, **the two disciplines this problem demands.**



Vivek Rao

CEO

Former private-equity investor and McKinsey consultant focused on P&C insurance.



Federico Reyes

CTO

Stanford AI researcher. Graph neural nets and relational data at Kumo AI, acquired by Nvidia.



Philo Bishay

CPO & CHIEF ACTUARY

Former chief actuary at W. R. Berkley E&S. Capital modeler at Gen Re; built and sold a brokerage.



Michael Palmich

HEAD OF SALES

25+ years in the actuarial software space, most recently at Akur8 and Milliman.



Eric Lam

FOUNDING ENGINEER

MIT for CS, physics, and math, with **three actuarial exams** done.

ADVISORY BOARD



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Joanne Chen

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Above all, we want to help you **win**.

We are a thought partner first. This technology is powerful, and it only helps once it is opinionated: built with a real point of view on how the work gets done, how it gets checked, and how it gets defended when a regulator asks.

We are here to push the frontier forward with the right partners, not to sell you a license. A law firm does the legal work rather than selling software to lawyers, and we hold ourselves to that standard: **we take ownership of the outcome, not just the tooling**. Every deployment starts with a proof of concept anchored on your own success metrics, and we work with your team to find where the workbench sets the shared standard first.

Let's set up a call

sales@tesora.ai

DE-RISKED BY DESIGN

- ✓ It runs in your own environment, in a dedicated tenant. No cross-tenant inference.
- ✓ Your rate plans, filings, and books are never used to train shared models.
- ✓ AES-256 at rest, TLS 1.3 in transit, and your own SSO.
- ✓ We clear your security review before any work touches a filing.



SOC 2 Type II. Independent auditors review us annually, and our trust center shows the continuous testing live. SOX-aligned against both actuarial and IT standards.

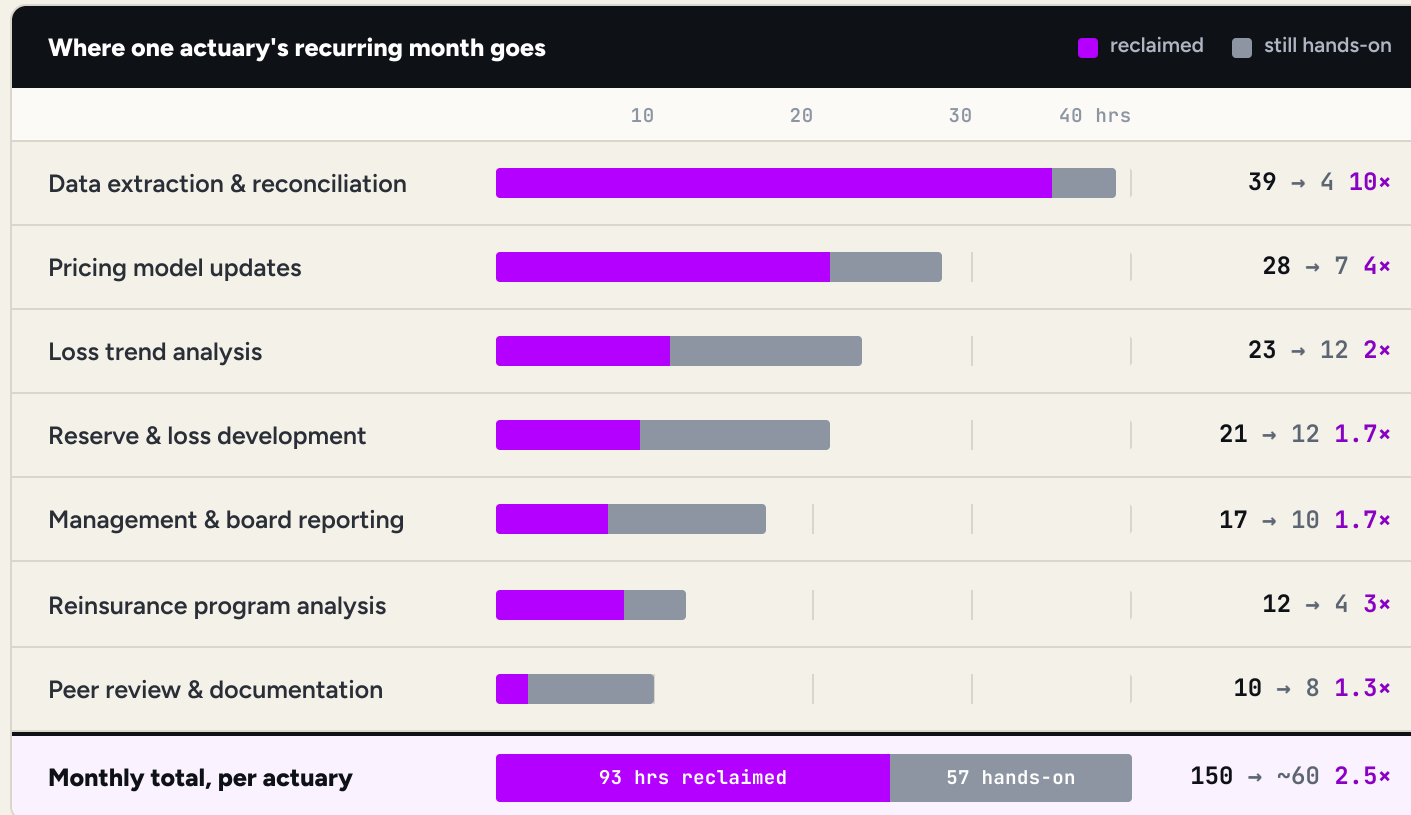
Appendix



Per actuary, the recurring month goes from **150 hours to about 60.**

Most of the month disappears into recurring work: the data pulls, the reconciliations, the model updates.

Tesora compresses that mechanical layer, so the hours come back as judgment rather than assembly.



~90 hours back per actuary, per month

A 2.5× blended throughput gain across the recurring month, with the same team and the same sign-off.

Smarter, not harder

Time freed for strategy and selections rather than assembly.

Unlock your actuaries

More products, more territories, and more actuarial capability from the same team.

Hours are per actuary and reflect a typical ~150 hour recurring workload within a roughly 160 hour month, with per-workflow gains from Tesora's engagement framework. Figures are modeled and confirmed against measured results during a build.



We run on top of Claude, and we are model agnostic.

The model is the engine, not the product. What an actuary needs around it is infrastructure: the data wired in, the workflow decided, the cost controlled, and the record kept. That is the part we build, and we build it for you.

We guide the model

A raw model answers anything, confidently. Ours is held to an actuarial method: fit frequency and severity separately, test against your own history, and stop to ask before selecting.

We wire the infrastructure

Your loss runs, policy systems, filings, and ledger are connected once and stay connected, so every study reads the same reconciled book. A chat window starts empty every time.

The workflow is opinionated

Reserving and pricing have a right order of operations, and we encode it. Chain ladder, Bornhuetter-Ferguson, and Cape Cod run side by side, so the selection stays your actuary's call.

We manage the cost

Running a real book through a frontier model naively is slow and expensive. Extraction runs on a small model, and the judgment calls go to the strongest one.



We use the best models. **You keep the value.**

Every firm is working out its AI spend internally right now. Our team of AI researchers, actuaries, and industry veterans knows what these systems cost, and how to optimize model spend per business outcome.



Right model, right step

Each step goes to the model that gets it right for the least spend. Extraction runs small, judgment runs strong, and none of that routing is your problem to manage.



Our incentive is the outcome, not the token count

A model vendor is never going to tell you to buy less of what it sells. Our platform has to deliver the result, so keeping model spend down is work we have to solve for ourselves rather than work we are merely free to do. **You are never left at the mercy of a rogue agent burning through tokens.**



Latest models, always up

Your workflows, data wiring, and audit trail live with us, so the strongest model arrives without a migration or a new vendor. A system built in-house on one provider stops when that provider does. We route across them, so **servicing an outage is our problem rather than your stopped quarter.**



Context mapping

We map your data, formats, and quirks against the work they feed, so we know intimately which model fits where. That map deepens the longer we work with your team, and it carries across every upgrade.



Audits matter more, not less

Better models take on more complex work, which raises rather than lowers the bar on proving it. Every value stays cited and reproducible, and **the advantage compounds for whoever moves first.**